CS : Computer Science and Information Technology
GATE Paper 2022 – Solutions

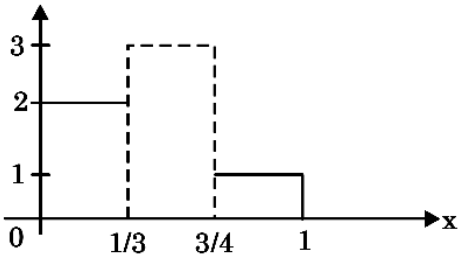
Answer Key

General Aptitude									
1.	(D)	**2.**	(C)	**3.**	(D)	**4.**	(A)	**5.**	(B)
6.	(B)	**7.**	(D)	**8.**	(D)	**9.**	(B)	**10.**	(D)
Computer Science and Information Technology									
11.	(A)	**12.**	(D)	**13.**	(D)	**14.**	(A)	**15.**	(A)
16.	(B)	**17.**	(A)	**18.**	(B)	**19.**	(C)	**20.**	(C)
21.	(D)	**22.**	(C)	**23.**	(B),(C),(D)	**24.**	(A),(B),(D)	**25.**	(A),(B)
26.	(A),(D)	**27.**	(A),(B),(C)	**28.**	509 to 509	**29.**	5 to 5	**30.**	36 to 36
31.	8 to 8	**32.**	7 to 7	**33.**	0.85 to 0.85	**34.**	–0.5 to –0.5	**35.**	4 to 4
36.	(A)	**37.**	(D)	**38.**	(C)	**39.**	(A)	**40.**	(C)
41.	(B)	**42.**	(D)	**43.**	(A)	**44.**	(A)	**45.**	(D)
46.	(A),(B),(C)	**47.**	(A),(B),(C)	**48.**	(B),(C),(D)	**49.**	(A),(B),(C)	**50.**	(A),(B),(C)
51.	(A),(B),(C)	**52.**	(A)	**53.**	(A),(C),(D)	**54.**	(A),(B),(D)	**55.**	(B),(D)
56.	2 to 2	**57.**	0.5 to 0.5	**58.**	24 to 24	**59.**	7.07 to 7.09	**60.**	33 to 33
61.	1.42 to 1.45	**62.**	0 to 0	**63.**	153 to 153	**64.**	0.6 to 0.6	**65.**	80 to 80

General Aptitude

1. (D)
The fare is too high for it to be considered fair.

2. (C)



$$\text{Area} = 2\times\frac{1}{3}+3\times\left(\frac{3}{4}-\frac{1}{3}\right)+1\times\left(1-\frac{3}{4}\right)$$

$$= \frac{2}{3}+3\times\frac{5}{12}+1\times\frac{1}{4}$$

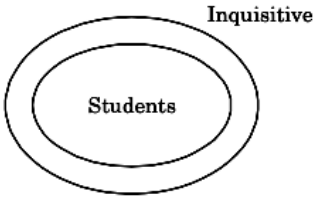
$$= \frac{2}{3}+\frac{15}{12}+\frac{1}{4}$$

$$= \frac{8+15+3}{12}$$

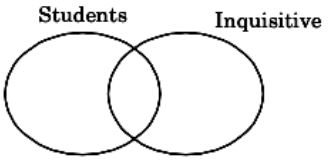
$$= \frac{26}{12}=\frac{13}{6}$$

3. (D)
r be the root of the equation $x^2 + 2x + 6 = 0$
so it will satisfy
$r^2 + 2r + 6 = 0$ ….(1)
Now, $(r + 2)(r + 3)(r + 4)(r + 5)$
$\Rightarrow$ $(r^2 + 5r + 6)(r^2 + 9r + 20)$
$\Rightarrow$ $(r^2 + 2r + 6 + 3r)(r^2 + 2r + 6 + 7r + 14)$
$\Rightarrow$ $(0 + 3r)(0 + 7r + 14)$
$\Rightarrow$ $3r(7r + 14)$
$\Rightarrow$ $21(r^2 + 2r) = 21(-6)$
$\Rightarrow$ -126

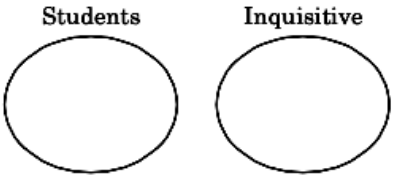
4. (A)



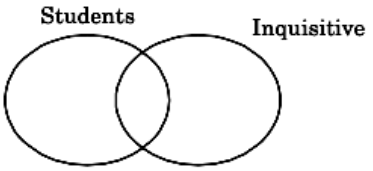
1. All students are inquisitive.



2. Some students are inquisitive.
   If all are true then some also true so first and second can be true simultaneously.



3. No student is inquisitive.
4. Some students are not acquisitive.



If some students are in inquisitive true then some students are not inquisitive is also true. Second and fourth can be true simultaneously.

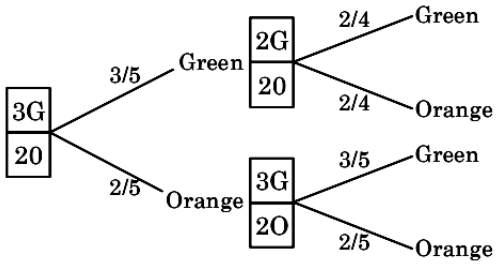
5. (B)
A word, sentence or a number that reads the same backward or forward.
Ex. RADAR
MADAM
REFER
12321
So, option (B) $\boxed{R}\boxed{A}\boxed{R}$ (after rotating 2nd and 3rd plates) is a palindrome.

6. (B)
Metrics are useful in organizations with good work culture

7. (D)
Let total candidates appeared = x
Appeared boys = 65% x = 0.65x
Appeared girls = 35% x = 0.35x
Let total qualified = y

qualified boys = 40% y = 0.4y
qualified girls = 0.65%y = 0.6y
Option (D) is correct because 0.6y > 0.4y.

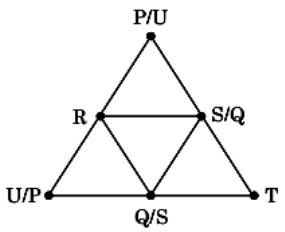
8. (D)
G → Green
O → Orange



$$P(E) = \frac{3}{5} \times \frac{2}{4} + \frac{2}{5} \times \frac{2}{5}$$

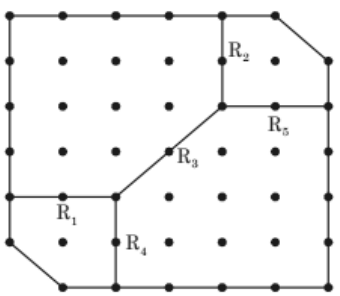
$$= \frac{3}{10} + \frac{4}{25} = \frac{23}{50}$$

9. (B)
Using above information we can draw,



S can't be place at a corners because PR is parallel with QS.

10. (D)



R_3 → First additional Rope
R_4 → Second additional Rope
R_5 → Third additional Rope
So, using 3 additional ropes. We are able to divide into 4 similar shape plots.

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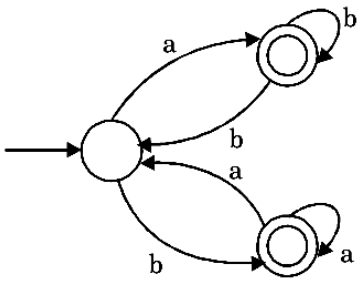
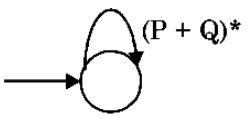
11. (A)
Growth rate of exponential function is greater than polynomial function.
Thus, $f(n^2) = \theta(f(n)^2)$, when f(n) is a polynomial.

12. (D)
Given finite automation,

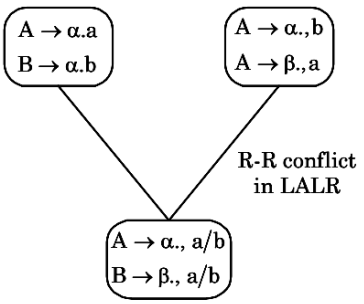




(P + Q)* (ab* + ba*)
(ab*b + ba*a) * (ab* + ba*)

13. (D)
We can create a LR parsser to pass a DCFL language. Sometime LR parser cannot parse context free language(CFL) because CFL are sometime inherently ambiguous and we cannot design a LR parser for ambiguous grammar. So, LR(1) is sufficient for DCFL.

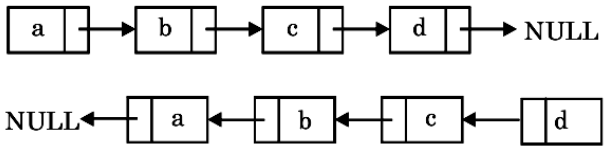
- Symbol table can be accessed in all phases of compiler.
- Data flow analysis is necessary in code–optimization.
- LALR may have reduce-reduce conflict even if CLR don't have any RR conflict (conflict arising due to merging of states where lookaheads are same).



14. (A)
At last one of the following holds in BCNF.
(i) $\alpha \rightarrow \beta$ is a trivial functional dependency i.e $\beta \subseteq \alpha$.
(ii) α is a superkey.

Thus, a relation with only two attribute must be in BCNF.
BCNF decomposition doesn't preserve functional dependencies.
It is not mandatory that every relation has at least one non-prime attribute.
If all attributes of relation are prime attribute, then the relation is always in 3NF.

15. (A)
Three pointers are required to reverse the linked list.
So, in worst case $\theta(n)$ time required to reverse the linked list.



16. (B)
Number of keys = n
Number of slots = m
As per question, we need to find out number of expected keys in a slot, i.e. each slot how many keys are possible.

So, each slot expected key should be $\frac{n}{m}$.

17. (A)
In programmed I/O CPU time is wastes.
In interrupt driven I/O at particular time interval CPU check the interrupt and CPU transfer the data to memory as per interrupt requirement. But it doesn't transfer the data with maximum throughput.
In polling based I/O, the I/O device is continuously poll by asking whether it needs CPU or not. So it also transfer data to main memory with low throughput.
DMA based I/O transfer doesn't involved CPU for transferring data to main memory and its throughput is maximum.

18. (B)
Given two four bit registers R_1 and R_2.
(A) $R_1 = 1011$
$R_2 = 1110$

$$\begin{array}{r} R_1 + R_2 = \\ \\ \\ \end{array} \begin{array}{r} 1\ 0\ 1\ 1 \\ 1\ 1\ 1\ 0 \\ \hline 1\ 0\ 0\ 1 \end{array}$$

No overflow occurred, because sign bit is same for $(R_1 + R_2)$
(B) $R_1 = 1100$
$R_2 = 1010$

$$\begin{array}{r} R_1 + R_2 = \\ \\ \\ \end{array} \begin{array}{r} 1\ 1\ 0\ 0 \\ 1\ 0\ 1\ 0 \\ \hline 0\ 1\ 1\ 0 \end{array}$$

Overflow occurred, because sign bit is different for $(R_1 + R_2)$.

(C) $R_1 = 0011$
$R_2 = 0100$

$$R_1 + R_2 = \begin{array}{r} 0\ 0\ 1\ 1 \\ 0\ 1\ 0\ 0 \\ \hline 0\ 1\ 1\ 1 \end{array}$$

No overflow occurred, because sign bit is same for $(R_1 + R_2)$.

(D) $R_1 = 1001$
$R_2 = 1111$

$$R_1 + R_2 = \begin{array}{r} 1\ 0\ 0\ 1 \\ 1\ 1\ 1\ 1 \\ \hline 1\ 0\ 0\ 0 \end{array}$$

No overflow occurred, because sign bit is same for $(R_1 + R_2)$.

19. (C)
Given threads are T_1, T_2 and T_3 and three binary semaphore variable is used for synchronization S_1, S_2 and S_3.

T_1	T_2	T_3
while (true) { wait (S_3); print (″C″) ; signal (S_2) ;	while (true) { wait (S_1); print (″B″) ; signal (S_3) ;	while (true) { wait (S_2); print (″A″) ; signal (S_1) ;

Given sequence need to print,
BCABCABCA...
First element in this sequence is 'B'. It means thread T_2 should execute first.
Thus, at this moment
$S_1 = 1$, $S_2 = 0$, $S_3 = 0$

20. (C)
The trace of a matrix is the sum of the diagonal elements of the matrix.

- In this question, property of trace is used that is trace of product (AB) = trace of product (BA)
- Statement I $\rightarrow$ it said tr $(A_{m\times n} \times B_{n\times m}) = tr(B_{n\times m} \times A_{m\times n})$

Let take example,

$$= \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2\times 3}, B = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}_{3\times 2}$$

$$tr(AB) = \begin{bmatrix} 14 & 22 \\ 32 & 77 \end{bmatrix} = 14 + 77 = 91$$

$$tr(BA) = \begin{bmatrix} 17 & 22 & 27 \\ 22 & 29 & 36 \\ 27 & 36 & 45 \end{bmatrix} = 17 + 19 + 45 = 91$$

$91 = 91$

Statement–I is correct.

Statement–II

$tr(C_{n\times n} \times D_{n\times n}) = tr(D_{n\times n} \times C_{n\times n})$

Take example,

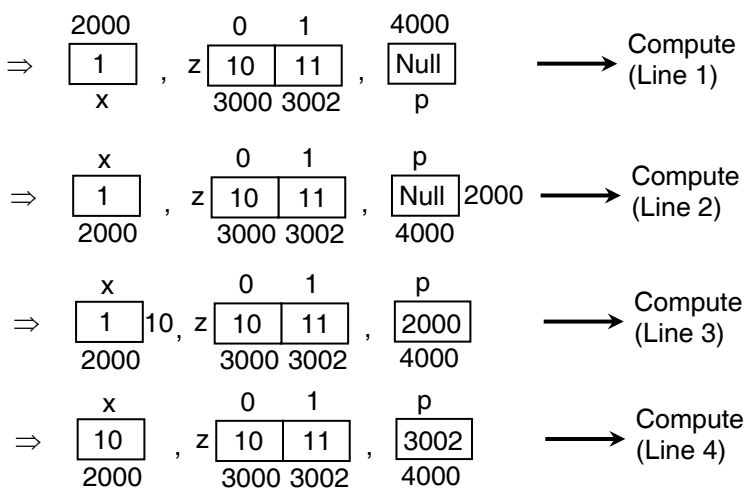
$$C = \begin{bmatrix} 2 & 2 \\ 3 & 5 \end{bmatrix}_{2\times 2}, D = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2\times 2}$$

$$tr(CD) = \begin{bmatrix} 8 & 12 \\ 18 & 26 \end{bmatrix} = 8 + 26 = 34$$

$$tr(DC) = \begin{bmatrix} 8 & 12 \\ 18 & 26 \end{bmatrix} = 26 + 8 = 34$$

Statement–II is also true.

21. (D)



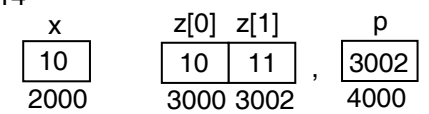
$\Rightarrow$ *(&z[0] + 1) + = 3

$\Rightarrow$ *(3000 + 1) + = 3

$\Rightarrow$ *(3002) + = 3

$\Rightarrow$ (11)+ = 3

$\Rightarrow$ 14



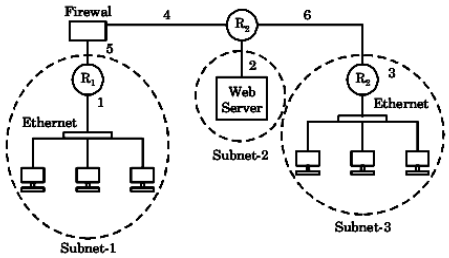
$\Rightarrow$ x = 10

z[0] = 10

z[1] = 14

22. (C)

Given an enterprise network with two Ethernet segments, a web server and a firewall connected via three routers. Each interface of router formed a subnet.



Number of subnet = 6

23. (B),(C),(D)

(A) It is not necessary that every subset of a recursively enummable language is recursive.

(B) If language L and its complement ($\overline{L}$) both are recursively enumerable then L must be recursive.

(C) Context-free language is recursive so its complement is also recursive.

(D) Regular languages are closed under intersection, so their intersection must be deterministic context-free.\

24. (A),(B),(D)

In write through policy, all the write operation is made in main memory and cache memory simultaneously, ensure that main memory is valid.

In write back policy, at the time of block replacement when dirty bit is set on the line changes is written back into the memory.

The cache eviction is a strategy in which the data is removed from the cache.

(i) To make room for more relevant cache entries.

(ii) To shrink the cache to make available more RAM for other users.

(A) In write back, dirty bit is set for those lines which are updated.
In write through, no dirty bit is required. Hence option (A) is FALSE.

(B) In write back, data transfer from cache to memory is required at the time of block replacement, i.e. when eviction required. Hence, option (B) is FALSE.

(C) In write through, no need to do eviction of a block from cache. So there is no data transfer required from cache to main memory. Hence, option (C) is TRUE.

(D) For read/write misses in write back, a line needed to be evicted for the newly fetched block. Hence, option (D) is false.

25. (A),(B)

Given three relation,

Employee(eId, Name)

Brand(bId), bName)

Own(eId, bId)

Need to find “set of eIds who own all the brands”.

Required answer,

$\pi_{eId}(\pi_{eId,bId}\ (own)\ /\ \pi_{bId}\ (Brand))$

$\pi_{eId}(own) - \pi_{eId}\ ((\pi_{eId}\ (own) \times \pi_{bId}\ (Brand)) - \pi_{eId,\ bId}\ (own))$

26. (A),(D)

(A) Circular wait is a necessary condition for the formation of deadlock.
(B) No deadlock occurred when more than one instance of resource exist.
(C) An unsafe state doesn't leads to deadlock.
(D) If there is no request edge then no deadlock.

27. (A),(B),(C)

(A) $(xy)^2 = x^2y^2$

$\Rightarrow$ xy xy = xx yy

Take x^{-1} and y^{-1} on both side.

x^{-1} xy xy y^{-1} = x^{-1} xx yyy^{-1}

yx = xy, so it is commutative.

(B) $\forall x \in G$, $x^2 = 1$ then G is

$xx = 1 \Rightarrow x = x^{-1}$

If every element has its own inverse in a graph, then graph is commutative.

(C) If the order of 'G' is 2 means it is order of prime, so 'G' is commutative.

(D) If G is commutative then the subgroup is G is also commutative.

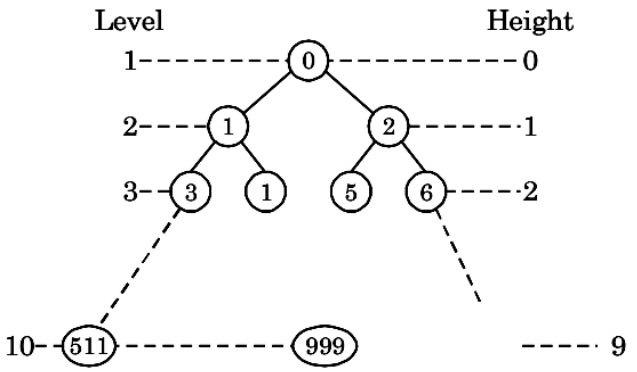
Let take I is a subgroup of G, if e, f $\in$ I, we have e, f $\in$ G, then ef = fe. So, I is commutative.

28. 509 to 509

Given binary search tree with 1000 distinct elements. And the tree is stored using the array representation of binary heap trees. Array indices start with 0.

Index number of first node at each level can be found by $(2^h - 1)$, where h is the height of the tree.

And, Number of node at each level can be find out by (2^{n-1}), where n is the number of level.



At 10th level number of nodes = 2^{10-1} = 512.

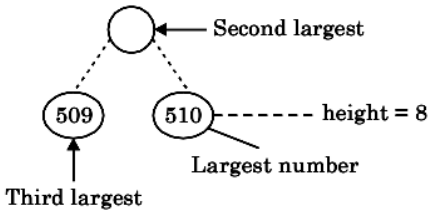
At height 9, index number of first node = 2^{9-1} = 511

Since, total number of node is 1000, so we need to check upper level. Because rightmost number in binary seachtree is maximum.

At 9th level, no. of nodes = 2^{9-1} = 256.

At high 8, index no. of first node = 2^{8-1} = 255 .

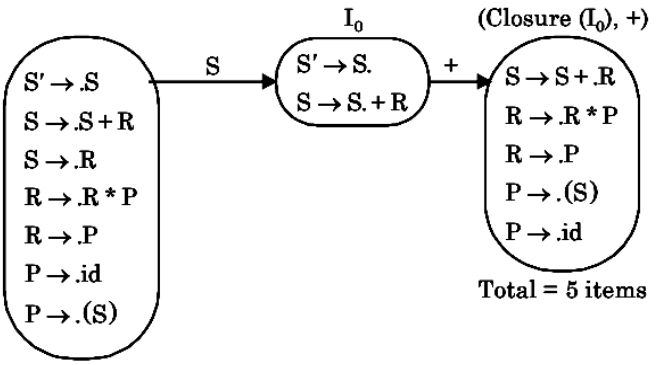
Index number of last node in 9th level, = 255 × 2 = 510



Required answer is 509.

29. 5 to 5

Apply augmented grammar is :



30. 36 to 36

We are given that vertices = n

The maximum number of possible edges in an undirected graph with 'n' vertices and 'k' components is :

Example :

To get maximum edges, take one vertex each for each complement, except last component.

Now, (K–1) components have 1 vertex each and last component has n–(K–1) vertices. Make the last component complete i.e. it has

$${}^{n-(k-1)}C_2 = \frac{(n-K)(n-K+1)}{2}$$

Number of edges (e) = $\frac{(n-K)(n-K+1)}{2}$

Complement (K) = 2

$$= \frac{(10-2)(10-2+1)}{2}$$

$$= \frac{8\times 9}{2} = 36$$

31. 8 to 8

Given three functional dependencies,

$AB \rightarrow C$

$BC \rightarrow D$

$C \rightarrow E$

Compute closure of attribute,
$(AB)^+ = \{A, B, C, D, E\}$
$(BC)^+ = \{B, C, D, E\}$
$C^+ = \{C, E\}$
Only one candidate key is possible for given relation : {A B}
Superkey can be found by adding any combination of attribute {C, D, E} with candidate key (AB).
Number of possible combination with candidate key required = $2^3 = 8$.

32. 7 to 7
So, balls and bins are identical, the possible ways are :
We have three bins.

B_1	B_2	B_3
6	0	0
5	1	0
4	1	1
4	2	0
3	2	1
3	3	0
2	2	2

Total ways is 7.

33. 0.85 to 0.85
Cache memory hit rate = 0.8
Access Latency = 10 nsec.
Miss penalty = 100 nsec.
Optimization increases cache access latency to 15 nsec.
Miss penalty is not affected.
Average memory access time in first case
$= 0.8 \times 10 + (1 - 0.8) \times 100 = 28$ n sec
Average memory access time in second case
$= x \times 15 + (1 - x) \times 100 = 28$
$15x + 100 - 100x = 28$
$85x = 72$
$$x = \frac{72}{85} = 0.85$$

34. –0.5 to –0.5
$$\lim_{x \to 0^+} \frac{\sqrt{x}}{1 - e^{2\sqrt{x}}}$$
$\Rightarrow$ put 0 in equation $\Rightarrow \frac{0}{1-1} = \frac{0}{0}$
Apply L' hospital rule, we get (differentiate numerator and denominator)

$$\Rightarrow \lim_{x\to 0^+} \frac{\frac{1}{2\sqrt{x}}}{0 - e^{2\sqrt{x}} \cdot \frac{2}{2\sqrt{x}}}$$

$$\Rightarrow \lim_{x\to 0^+} \frac{\frac{1}{2}\sqrt{x}}{-2e^{2\sqrt{x}} \cdot \frac{1}{2\sqrt{x}}}$$

$$\Rightarrow \lim_{x\to 0^+} \frac{1}{-2e^{2\sqrt{x}}}$$

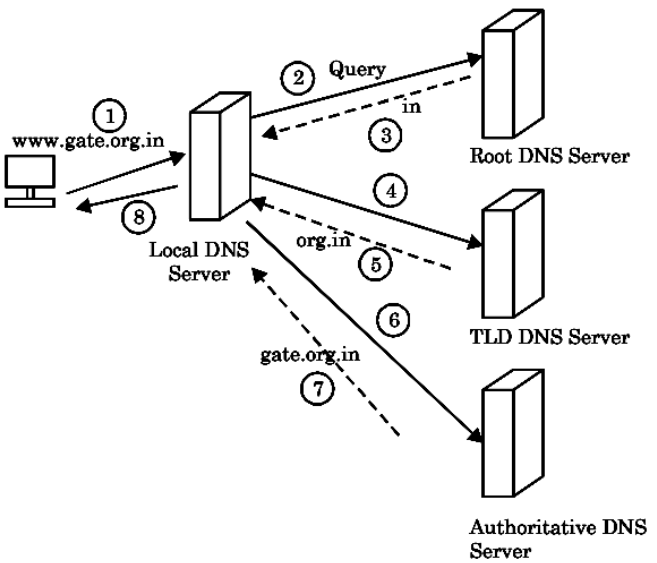
$$\Rightarrow -\frac{1}{2} = -0.5$$

35. 4 to 4

Given domain name,

www.gate.org.in

Iterative query mechanism is used in the resolution,



There are 4-pairs involved in completely resolving the domain name.

36. (A)

$$a_n = \begin{cases} n+1, & n \text{ is odd} \\ 1, & \text{otherwise} \end{cases}$$

$$\frac{(n-1)(n-K+1)}{2} = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + ...$$

$$= 1 + 2x + x^2 + 4x^3 + x^4 + 6x^5 + ...$$

$$= (1 + x + x^2 + x^3 + ...) + x(1 + 3x^2 + 5x^4 + ...)$$

$$= \frac{1}{(1-x)} + x\frac{d}{dx}(x + x^3 + x^5 + \ldots)$$

$$= \frac{1}{(1-x)} + x\frac{d}{dx}\left[\frac{x}{(1-x^2)}\right]$$

$$= \frac{1}{(1-x)} + \frac{x(1+x^2)}{(1-x^2)^2}$$

37. (D)

All pair shortest paths for adjacency matrix = A_{ij}^n.

A_{ij}^n = n vertices can walk from i – j.

Since, the cycle has 3 vertices and it is counted for every vertex, we need to divide by 3. To get the three vertex loop for directed graph = $\frac{A_{ij}^3}{3}$.

For undirected graph, A-B-C-A is same as A-C-B-A cycle. So, 2 possibility will be formed.

The number of 3-cycle for undirected graph is $\frac{A_{ij}^3}{3*2} = \frac{A_{ij}^3}{6}$.

38. (C)

(A) TLB perform parallel search.

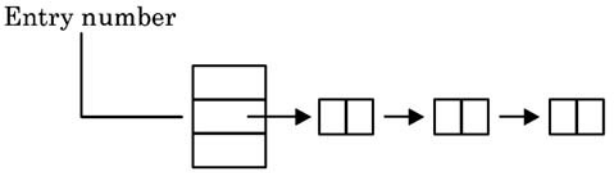
(B) TLB hit means word will always present in main memory.

(C) Inverted Page Table

Frame 0	Process	Page
Frame 1	Process	Page
	⋮	⋮

Memory access time using inverted page table is not some because in inverted page table no indexing is applied and there is no equal linear searching. So, this statement is false.

(D) Virtual address ⇒ entry number.

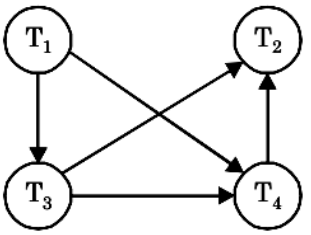


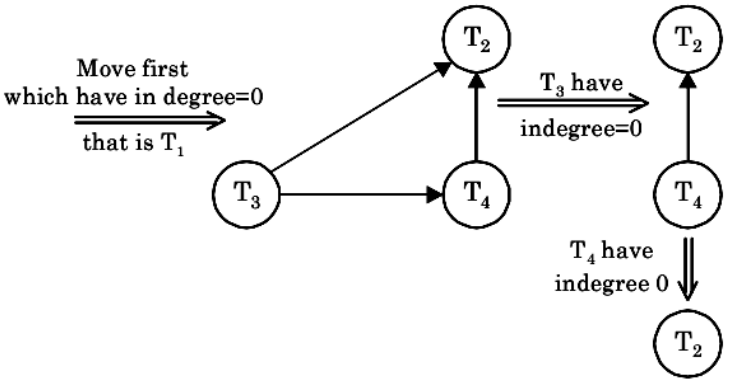
If they map to same value while hashing, there memory access time of addresses will not same because there is a chance that some elements are present at the end of linked list.

39. (A)

Given schedule S with four transactions,

T_1	T_2	T_3	T_4
			$R_4(x)$
	$R_2(x)$		
		$R_3(x)$	
$R_1(y)$			
$W_1(y)$			
	$W_2(x)$		
		$W_3(y)$	
			$R_4(y)$





Thus, $T_1 \rightarrow T_3 \rightarrow T_4 \rightarrow T_2$

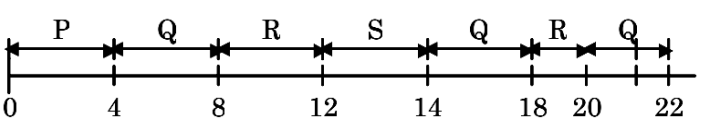
40. (C)

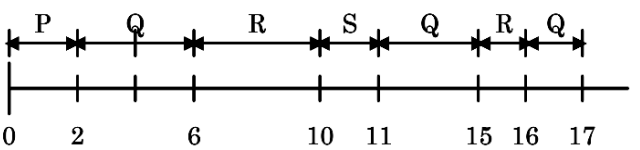
Given digital display system, in which 16-bit code word is used.
S is a 1024-word memory segment.
R is a 32-word register file.
So, there are 10-input required for decoder P and output of decoder P is 2^{10}.
Similarly, there are 5-input required for decoder Q and output of decoder Q is 2^5.
And T is a multiplexer, which takes 2 input and gives 1 output.

$P \equiv 10:2^{10}$ decoder
$R \equiv 5:2^{5}$ decoder
$T \equiv 2:1$ Multiplexer

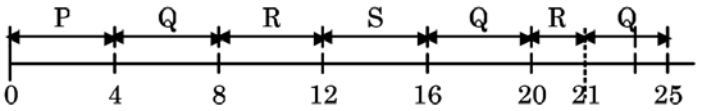
41. (B)
$C = A + B$
$R_A = 0 \times C1400000 \Rightarrow 1100\ 0001\ 0100\ 00000$
$R_B = 0 \times 42100000 \Rightarrow 0100\ 0010\ 0001\ 00000$
$R_C = 0 \times 41400000 \Rightarrow 0100\ 0001\ 0100\ 00000$
If you see carefully A and C are changes only in sign bit.
IEEE–754 single precision = $(-1)^S * 1.M * 2^{E-127}$.
In every register, first bit is sign bit, next 8 bit is for exponent, and after that remaining bit is mantissa.
R_A = 1100 0001 01000 00000
= Sign = (–)
= Exponent = 130
$= (-1)^{-1}\ 1.1 \times 2^{130-127}$
$= -1.1 \times 2^3$
$= (-1100)_2 = -12$
R_C = Similarly $R_C = +12$ because only difference is sign bit.
R_B = 0100 0010 0001 00000
= Sign = (+)
= Exponent = 132
$= +1.001 \times 2^{132-127}$
$= +1.001 \times 2^5$
= +(100100) = +36
So, false option is C = A + B

42. (D)
Given four processes P, Q, R and S.
Scheduling algorithm : Round Robin
Time quanta = 4 time unit.
All the processes arrives at time t = 0.
(i) Exactly one context switch from S to Q.
(ii) Exactly one context switch from R to Q.
(iii)Exactly two context switch from Q to R.
(iv) No context switch from S to P.
(v) Switching to a ready process after the termination of another process is also considered a context switch.

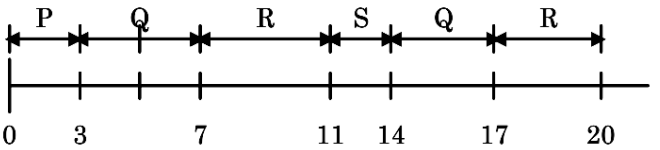
(A)


(B)


(C)

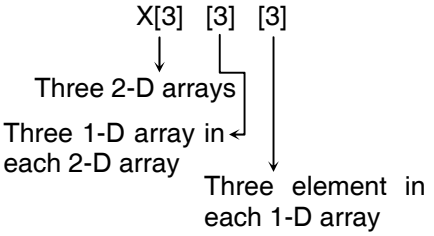


(D)



In option (D), there is no context switching exist from R to Q.

43. (A)



a[0]

	0	1	2
0	1	2	3
1	4	5	6
2	7	8	9

a[1]

	0	1	2
0	10	11	12
1	13	14	15
2	16	17	18

a[2]

	0	1	2
0	19	20	21
1	22	23	24
2	25	26	27

a[000] = 1
a[001] = 2
a[002] = 3
a[100] = 10
a[101] = 11
a[102] = 12
a[200] = 19
a[201] = 20
a[202] = 21

44. (A)

Step-I $\Rightarrow$ $\underset{a}{\boxed{80}}$ $\underset{b}{\boxed{120}}$ because we take small 'p' and capital 'X'.

char c $\Rightarrow$ (a & b) + '*'

$\Rightarrow$ Bitwise & = (80 & 120) + 42

$\Rightarrow$ 80 = 1010000

120 = 1111000

a&b = 1010000 = 80 + 42 = 12 = z

char d $\Rightarrow$ (a | b) + '–'

$\Rightarrow$ Bitwise | = (80 | 120) + 45

$\Rightarrow$ 80 = 1010000

120 = 1111000

a | b = 1111000 = 120 – 45 = 75 = K

Char e $\Rightarrow$ (a ^ b) + '+'

$\Rightarrow$ Bitwise ^= (80 ^ 120) + 43

$\Rightarrow$ a ^ b = 1010000

= 1111000

a ^ b = 0101000 = 40 + 43 = 83 = S

Print character = (z, K, S)

45. (D)

Let take coefficient matrix,

$$A = \begin{bmatrix} 1 & 1 & -2 \\ 1 & 3 & -1 \\ 2 & 1 & -5 \end{bmatrix}$$

Perform row-column operation.

$R_2 \to R_2 - R_1$

$$\begin{bmatrix} 1 & 1 & -2 \\ 0 & 2 & 1 \\ 2 & 1 & -5 \end{bmatrix}$$

$R_3 \to R_3 - 2R_1$

$$\begin{bmatrix} 1 & 1 & -2 \\ 0 & 2 & 1 \\ 0 & -1 & -1 \end{bmatrix}$$

$$R_3 \to R_3 - \left(-\frac{1}{2}\right)R_2$$

$$u = \begin{bmatrix} 1 & 1 & -2 \\ 0 & 2 & 1 \\ 0 & 0 & -\frac{1}{2} \end{bmatrix}$$

$A = L * u$

$$\begin{bmatrix} 1 & 1 & -2 \\ 1 & 3 & -1 \\ 2 & 1 & -5 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 2 & -\frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 2 & 1 \\ 0 & 0 & -\frac{1}{2} \end{bmatrix}$$

$u_{33} = -\frac{1}{2}$

$L_{32} = -\frac{1}{2}$

46. (A),(B),(C)
(A) Undecidable
(B) Undecidable
(C) Undecidable
(D) Decidable, Turing Machine M decide if language L takes more than 1073 steps. Language $\overline{L}$ takes almost 1073 steps. So, it is decidable. Then its complement is also decidable. Hence, L is decidable.

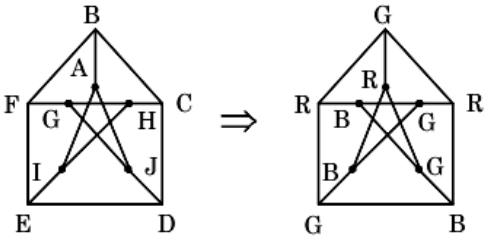
47. (A),(B),(C)
Given language,
$L_1 = \{a^n w a^n \mid w \in \{a,b\}^*\}$
$L_2 = \{wxw^R \mid w, x \in \{a,b)^*, |w|, |x| > 0\}$
Regular expression can be written for the language L_1 and L_2.
For L_1,
$(a + b)*$
For L_2,
$a(a + b)^+ a + b(a + b)^+ b$
Hence, both language are regular. And regular language are context free.

48. (B),(C),(D)
Option (A) : L_1 is not context-free but L_2 and L_3 are deterministic context-free.
True, L_1 is not context-free and L_2 and L_3 are clearly DCFLs since they have only one comparison and DPDA can accept both.
Option (B) : Neither L_1 nor L_2 is context-free.
False, L_2 is DCFL and every DCFL is a CFL.
L_2 – Context-free
L_3 – Context-free
$L_2 \cap L_3 = \{a^n b^n c^n \text{ or } a^m b^m c^m, m, n \geq 0\}$
This language is context sensitive language.
Option (C) : L_2, L_3 and $L_2 L_3$ all are context-free
False,
L_2 – Context-free
L_3 – Context-free
$L_2 \cap L_3 = \{a^n b^n c^n, n \geq 0\}$. We cannot compare the $a^n b^n c^n$ with one single stack.
Hence it is not context-free.

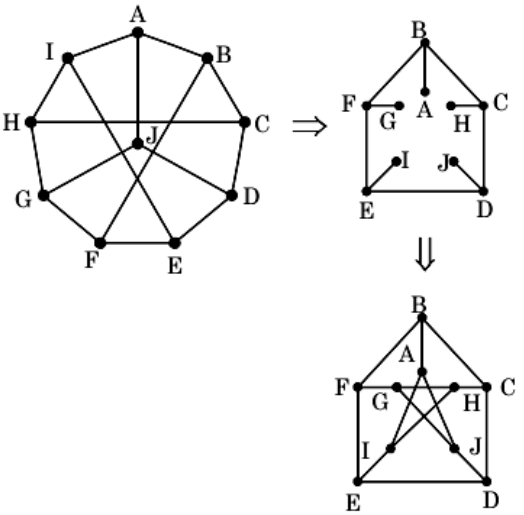
Option (D) : Neither L_1 nor its complement is context-free.
False, L_1 – Not context-free
L_1 Complement of L_1 is accepted by a context-free grammar is CFL.
Hence the correct answer is option (B), (C), (D).

49. (A),(B),(C)
(A) : True
(B) : True
(C) : True
(D) : No, G cannot have multiple minimum spanning trees when all the edges are distinct.

50. (A),(B),(C)
(A) Yes, it is true.



(B) Peterson graph has Hamiltonian path but not Hamiltonian cycle.
(C) Yes, it is isomorphic to the given graph.
Step-I



(D) Vertex independent set is a set of vertices which are not adjacent.
Maximum vertex independent set is a set in which we cannot add one more vertex to it. So, largest independent set of Peterson graph is 4.

51. (A),(B),(C)

Eliminating option is the way for solving this question ⇒ f(2n) is even function and f(2n + 1) is add function.

f(1) = 1

f(2) = f(2*1) = 2f(1) − 1 = 2 − 1 = 1 [f(2n) = 2f(n) − 1]

f(3) = f(2*2 + 1) = 2*f(1) + 1 = 3 [f(2n + 1) = 2(n) + 1]

f(4) = f(2*2) = 2f(2) − 1 = 2*1 − 1] 1

f(5) = f(2*2 + 1) = 2f(2) + 1 = 2*1 + 1 = 3

f(6) = f(2*3) = 2f(3) − 1 = 2*3 − 1 = 5

f(7) = f(2*3 + 1) = 2f(3) + 1 = 7

f(8) = f(2^3) = f(2*4) = 2f(4) − 1 = 2*1 − 1 = 1

f(9) = f(2*4 + 1) = 2f(4) + 1 = 2*1 + 1 = 3

f(10) = f(5*2) = 2f(5) − 1 = 2*3 − 1 = 5

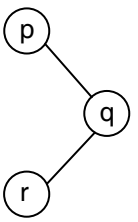
⋮

f(20) = f(5*4) = f(10*2) = 2f(10) − 1 = 2*5 − 1 = 9

- Computation of f(2), f(4) and f(8) show that f(2^n) = 1 is correct.
- Computation of f(3) and f(7) show that f(2^n − 1) = 2^n − 1 is correct.
- Computation of f(10) and f(20) show that f(5.2^n) = 2^{n+1} + 1 is correct.
- Computation of f(5) and f(9) show that f(2^n + 1) = 2^n + 1 is not correct.

52. (A)

Consider a graph ⇒



$$\text{K is adjacency matrix} = \begin{array}{c} \\ p \\ q \\ r \end{array}\begin{array}{c} \begin{array}{ccc} p & q & r \end{array} \\ \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \end{array}$$

$$K^2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Degree sequence of (p, q, r) = (1, 2, 1)

Diagonal elements= (1, 2, 1)

53. (A),(C),(D)

Use AX = λX in each options.

λ = Scalar quantity.

(A) AX = λX

$$\begin{bmatrix} -9 & -6 & -2 & -4 \\ -8 & -6 & -3 & -1 \\ 20 & 15 & 8 & 5 \\ 32 & 21 & 7 & 12 \end{bmatrix}\begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix} \Rightarrow AX = 1X$$

(B) AX = λX

$$\begin{bmatrix} -9 & -6 & -2 & -4 \\ -8 & -6 & -3 & -1 \\ 20 & 15 & 8 & 5 \\ 32 & 21 & 7 & 12 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 7 \\ -5 \\ 12 \\ 25 \end{bmatrix} \Rightarrow AX \neq \lambda X$$

(C) AX = λX

$$\begin{bmatrix} -9 & -6 & -2 & -4 \\ -8 & -6 & -3 & -1 \\ 20 & 15 & 8 & 5 \\ 32 & 21 & 7 & 12 \end{bmatrix}\begin{bmatrix} -1 \\ 0 \\ 2 \\ 2 \end{bmatrix} = 3\begin{bmatrix} -1 \\ 0 \\ 2 \\ 2 \end{bmatrix} \Rightarrow AX = 3X$$

(D) AX = λX

$$\begin{bmatrix} -9 & -6 & -2 & -4 \\ -8 & -6 & -3 & -1 \\ 20 & 15 & 8 & 5 \\ 32 & 21 & 7 & 12 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \\ -3 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 9 \\ 0 \end{bmatrix} = 3\begin{bmatrix} 0 \\ 1 \\ -3 \\ 0 \end{bmatrix} \Rightarrow AX = 3X$$

54. (A),(B),(D)

Cache Memory Size = 2 KB

Main Memory Size = 64 KB

Block Size = 64 B

$$\text{Number of Lines} = \frac{\text{Cache Memory Size}}{\text{Block Size}}$$

$$= \frac{2K}{64} = \frac{2^{11}}{2^{6}} = 2^{5}$$

16 bit

tag	Number of lines	Word Offset
5 bit	5 bit	6 bit

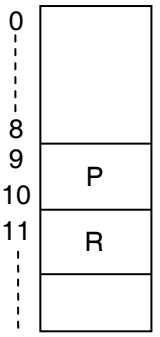
P:(A248)H 1010 0010 0100 1000; 9th block

Q:(C284)H 1100 0010 1000 0100; 10th block

R:(CA8A)H 1100 1010 1000 1010; 10th block

S:(A262)H 1010 0010 0110 0010; 9th block

But P&S are from same memory block (10100).



1st	2nd
P : M	P : Hit
Q : M	Q : M
R : M	R : M
S : Hit	S : Hit

- Every access of S is hit.
- Once P is brought to the cache it is never evicted.
- Every access to R evicts Q from cache.

55. (B),(D)

Given routing table,

1. 12.20.164.0 255.255.252.0
   12.20.10100100.00000000
2. 12.20.170.0 255.255.254.0
   12.20.10101010.00000000
3. 12.20.168.0 255.255.254.0
   12.20.10101000.00000000
4. 12.20.166.0 255.255.254.0
   12.20.10100110.00000000

Subnet (2) and (3) can be aggregated.

12.20.10101010.00000000
$\underbrace{12.20.10101000}_{\text{Network id}}\underbrace{.00000000}_{\text{Host id}}$

12.20.168.0/22

Subnet (1) and (4) can be aggregated

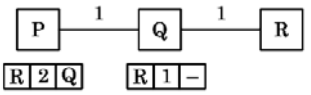
12.20.10100100.00000000
$\underbrace{12.20.10100110}_{\text{Network id}}\underbrace{.00000000}_{\text{Host id}}$

12.20.164.0/22

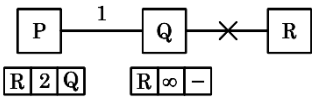
56. 2 to 2

Given schemes,
Student(SNO, SName, dNo)
Dept(dNo, dName)
Course(CNO, CName, dNo)
Register (SNo, CNo)
Select * from student as S where NOT EXIST
(select CNO from course where dNo = "DO1"
EXCEPT
Select CNO FROM Register WHERE SNO=S.SNO)
In above case, first inner query return is independent and return <{C11, C12>.
Second inner query is dependent on student relation. The condition is sNo = S.sNo.
So, for sNo = S01, second inner query returns <C11, C12>. And the except operation between two inner query gives empty result.
Except operation : R_1 EXCEPT $R_2 = R_1 - (R_1 \cap R_2)$
Now, NOT EXIST returns TRUE and outer query gives <S01, James, D01> as output.
When second inner query is checked for sNo = S02. It return <C11>. And the except operation between two inner query gives non-empty result. AND NOT EXISTS gives FALSE result. And outer query doesn't give any result.
Similarly for sNo = S04, outer query gives output <S04, jane, D01>.
Hence 2 rows found.

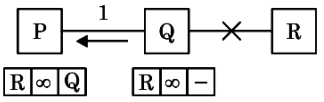
57. 0.5 to 0.5



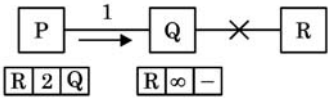
When link between Q and R is broken.

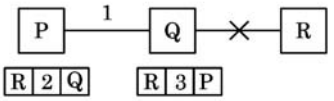


When Node P receives triggered update first from Q, then P successfully updated.



In this case no issue observed and count-infinity problem doesn't arised.
If node Q receives update first from node P.





So, routing table of Q is updated and it causes count-infinity problem.
So, out of two node, one node is responsible for count-to-infinity issue.
So, the probability of a routing loop formation between P and Q is 0.5.

58. 24 to 24

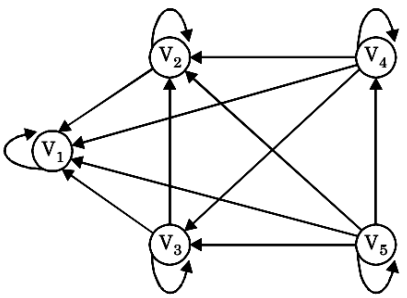
Given vertex,

$V = \{1, 2, 3, 4, 5\}$

A matrix 'A' is given in such a way that,

$$A[i][j] = \begin{cases} 1 & 1 \le j \le i \le 5 \\ 0 & \text{Otherwise} \end{cases}$$

$$\begin{array}{c|ccccc} & & & i \longrightarrow & & \\ & 1 & 2 & 3 & 4 & 5 \\ \hline 1 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 0 & 0 & 0 \\ j\downarrow 3 & 1 & 1 & 1 & 0 & 0 \\ 4 & 1 & 1 & 1 & 1 & 0 \\ 5 & 1 & 1 & 1 & 1 & 1 \end{array}$$



Directed spanning tree rooted at vertex 5 is $4 \times 3 \times 2 \times 1 = 24$

59. 7.07 to 7.09

Given, Bandwidth = 100 Mbps

Distance = 2100 Km.

Propagation speed = 3×10^8 m/sec

$$\text{Time taken to receive the packet} = \frac{\text{Packet size}}{\text{Bandwidth}} = \frac{1000\,\text{bytes}}{100\,\text{Mbps}}$$

$$= \frac{1000 \times 8\,\text{bits}}{100 \times 10^6\ \text{bits / sec}}$$

$= 0.08$ Msec.

$$\text{Propagation time} = \frac{\text{Distance}}{\text{Propagation speed}}$$

$$= \frac{2100\,\text{Km}}{3 \times 10^8\ \text{m / sec}}$$

$$= \frac{2100 \times 10^3\ \text{m}}{3 \times 10^8\ \text{m / sec}}$$

$= 7 \times 10^{-3}$ sec

$= 7$ m sec.

Required time = 7 + 0.08 = 7.08 m sec

60. 33 to 33

Bandwidth = 1 GbPS = $\frac{2^{30}}{8}$ bytes/sec

Maximum segment lifetime (MSL) = 60 seconds

Wrap around time = $\frac{\text{Total sequence number}}{\text{Bandwidth}}$

Wrap around time > Maximum segment lifetime

Total sequence $\frac{\text{Number}}{\text{Bandwidth}}$ > Maximum segment life time.

Suppose, minimum number of bits required for the sequence number field = x.

$$\frac{2^x}{1\times\frac{2^{30}}{8}} > 60$$

$$2^x = \frac{60\times 2^{30}}{8}$$

$$\log 2^x > \log\left(\frac{60\times 2^{30}}{8}\right)$$

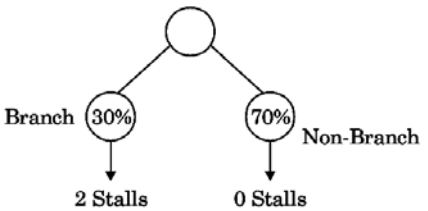
x > log 60 + 30 − log 8

Minimum value of x = 30 + 5.9 − 3 ≈ 33

61. 1.42 to 1.45

Cycle time (t_p) = 2 GHz = 0.5 nanoseconds

K = 5, $X_1 \Rightarrow$ without branch prediction.



CPI = 1 + Number of stalls instruction = 1 + (0.3) × 2 = 1.6

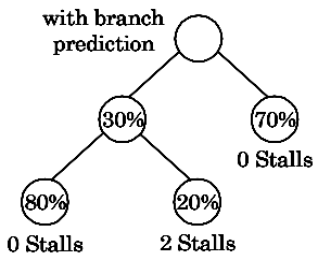
Average instruction execution time

= CPI ∗ Cycle time

= 1.6 ∗ 0.5 n sec

= 0.8 ns.

$X_2 \Rightarrow$ "If BPU predicted correct branch then it eliminate stalls but if BPU predicted wrong branch then BPU not add any additional stalls but remaining stalls we present."



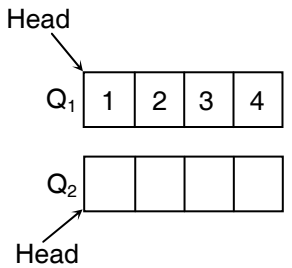
CPI = 1 + (0.3 × 0.2 × 2)
= 1.12

Average instruction execution time = CPI × Cycle time
= 1.12 × 0.5 ns = 0.56 ns

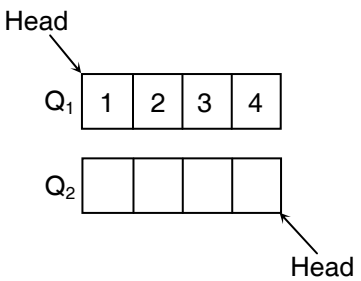
Speed up = $\frac{0.8}{0.56} = 1.42$

62. 0 to 0

Given queues Q_1 and Q_2

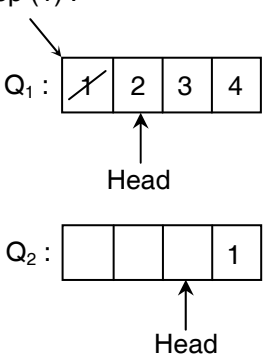


Traverse head of Q_2 at the end of queue so Queue format becomes,

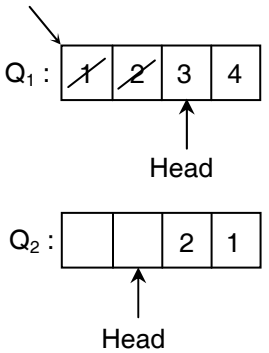


Now perform dequeue operation in Q_1 at head position and perform enqueue operation in Q_2 at head position.

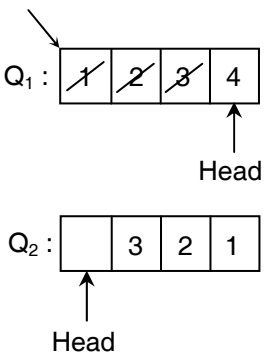
Step (1) :



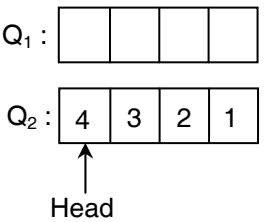
Step (2) :



Step (3) :



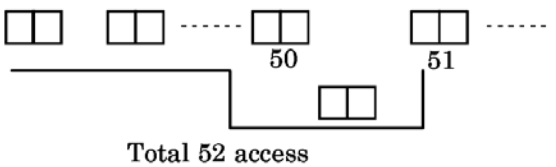
Step (4) :



Thus, there are zero enque operation performed on queue Q_1.

63. 153 to 153

For linked allocation, we need to access the block from 1 to 50, new block access that should insert between 50 and 51 block, and access 51 block. So, the total access is 52.



For contiguous allocation, we have 100 block, in contiguous allocation first we push the 100th block to 101th block (we need two access), then we push 99th block to

100th block (we need two access again) and so on upto 51th block. Now, 51 block is empty, then we push new block to 51 position list need one access). So, total access are

$$\underbrace{50\times 2}_{\text{(For 51 to 100 block)}} + \underbrace{1}_{\text{for new block}} = 100+1=101$$

Total required access are 101 + 52 = 153.

64. 0.6 to 0.6

Given page reference string,

7, 2, 7, 3, 2, 5, 3, 4, 6, 7, 7, 1, 5, 6, 1

$$\text{Page fault rate} = \frac{\text{Number of page faults}}{\text{Number of memory access}}$$

There are four page frames are given and LRU page replacement policy is used.

7	2	7	3	2	5	3	4	6	7	7	1	5	6	1
					5		5	5	7		7	7		
			3		3		3	3	3		1	1		
	2		2		2		2	6	6		6	6		
7	7		7		7		4	4	4		4	5		

Number of page fault = 9

Number of memory access = 15

$$\text{Page fault rate} = \frac{9}{15} = 0.6$$

65. 80 to 80

$S \rightarrow S_1 \# T \quad \{S.val = S_1.val * T.val\}$

$S \rightarrow T \quad \{S.val = T.val\}$

$T \rightarrow T_1 \% R \quad \{T.val = T_1.val \div R.val\}$

$T \rightarrow R \quad \{T.val = R.val\}$

$R \rightarrow id \quad \{R.val = id.val\}$

$\Rightarrow \; 20 \times 10 \div 5 \times 8 \div 2 \div 2$

Rule 1 : has higher priority then * because it is away from starting symbol.

Rule 2 : Both (÷ and *) are left associative.

$= 20 \times 10 \div 5 \times 8 \div 2 \div 2$

$= 20 \times 2 \times 4 \div 2$

$= 20 \times 2 \times 2$

$= 40 \times 2$

$= 80$

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